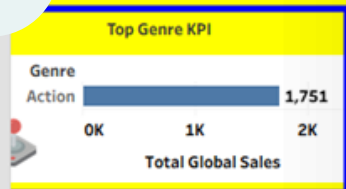
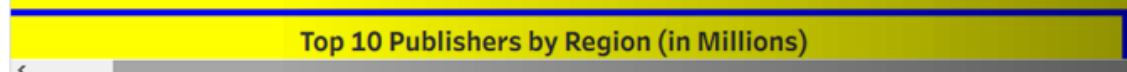
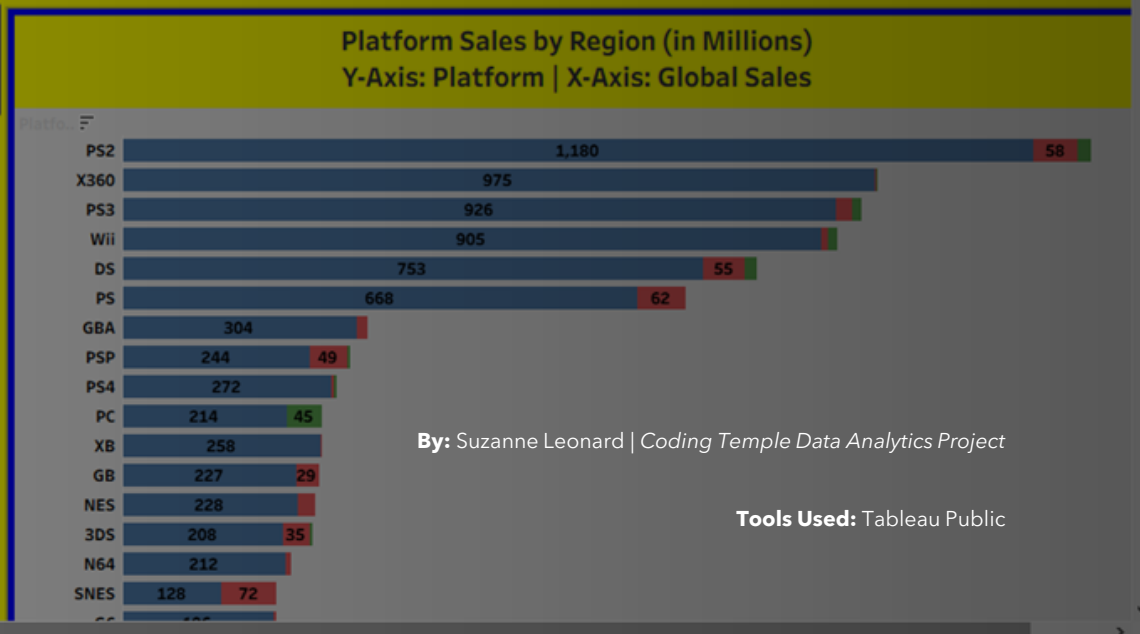
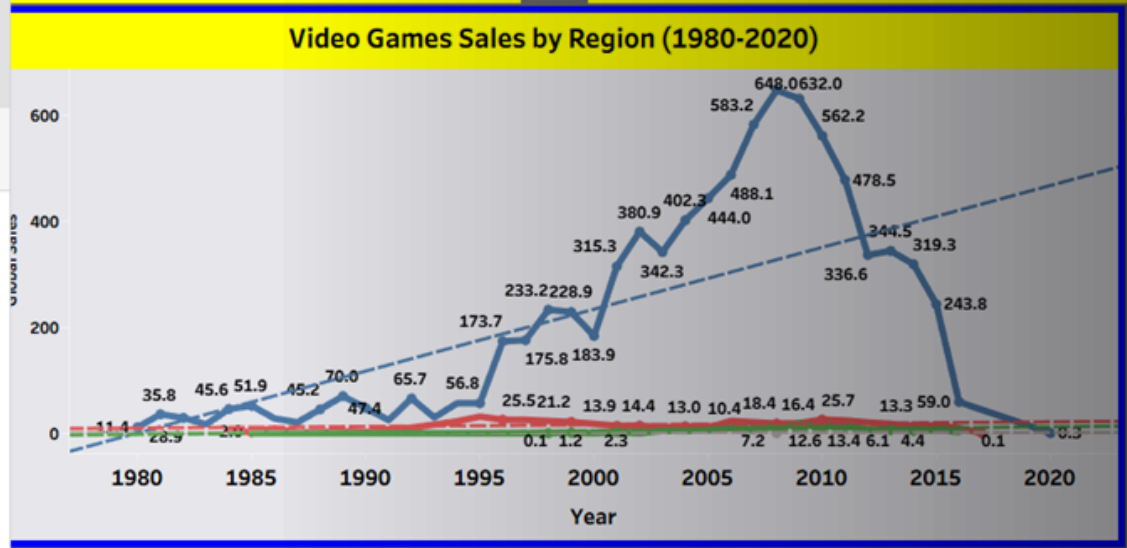


Title: Global Video Game Sales Analysis (1980-2020)



Insights (1980-2020):

- North America leads with 8,210M sales
- Top Genre: Action (1,723M)
- Top Platform: PS2 (1,233M)



By: Suzanne Leonard | Coding Temple Data Analytics Project

Tools Used: Tableau Public



Agenda

➤ **Sections:**

- Context & Purpose
- Executive Summary
- Key Findings
- Visualization Highlights
- Policy & Market Recommendations
- Methodology & Data Preparation
- Technical Documentation
- Future Outlook



Relate tables

Drag tables near vgsales.csv to relate them



Add a base table

Drag out additional base tables for multi-fact analysis.

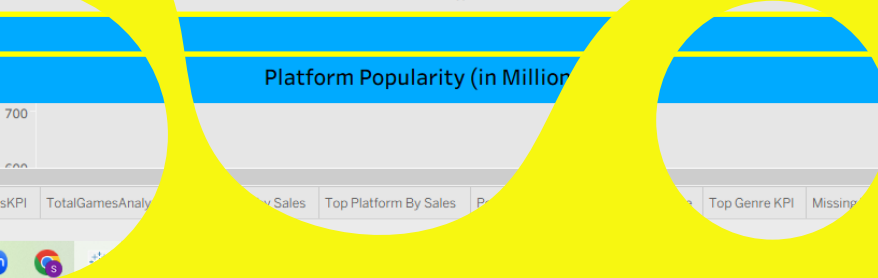
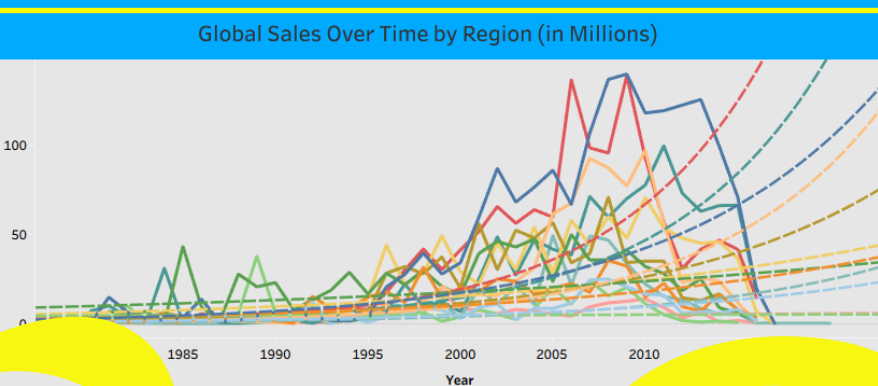
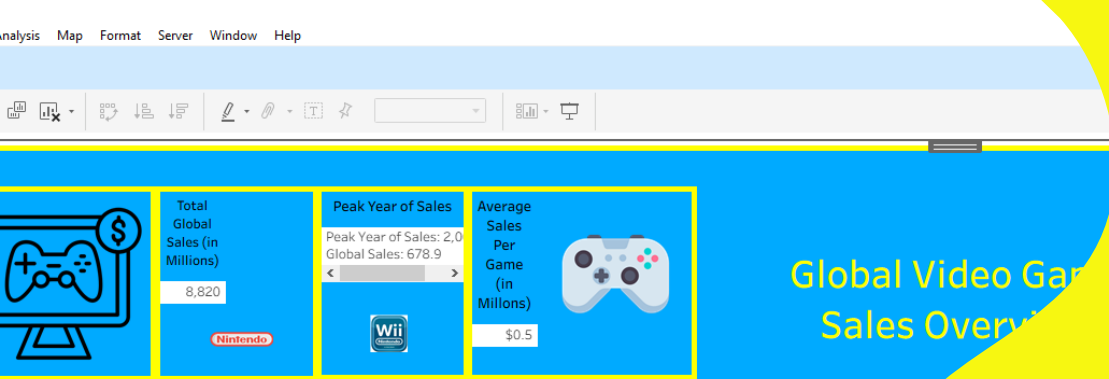
Need help adding data? [Learn more](#)

Context & Purpose

- The dataset from Kaggle and VGChartz includes 16,500+ games and global sales data (in millions) across **North America, Europe, Japan, and other regions**.
- **Purpose:**
- Identify long-term trends in genre and platform sales.
- Examine shifts in regional demand and publisher performance.
- Deliver practical recommendations for **publishers, developers, marketers**, and **regulators** based on real data trends.

#	Abc	Abc	#	Abc	Abc	
vgsales.csv	vgsales.csv	vgsales.csv	vgsales.csv	vgsales.csv	vgsales.csv	
Rank	Name	Platform	Year	Genre	Publisher	
1	Wii Sports	Wii	2006	Sports	Nintendo	
2	Super Mario Bros.	NES	1985	Platform	Nintendo	
3	Mario Kart Wii	Wii	2008	Racing	Nintendo	15.0000
4	Wii Sports Resort	Wii	2009	Sports	Nintendo	15.7500
5	Pokemon Red/Pokemon Blue	GB	1996	Role-Playing	Nintendo	11.2700
6	Tetris	GB	1989	Puzzle	Nintendo	23.2000
7	New Super Mario Bros.	DS	2006	Platform	Nintendo	11.3800
8	Wii Play	Wii	2006	Misc	Nintendo	14.0300
9	New Super Mario Bros. Wii	Wii	2009	Platform	Nintendo	14.5900

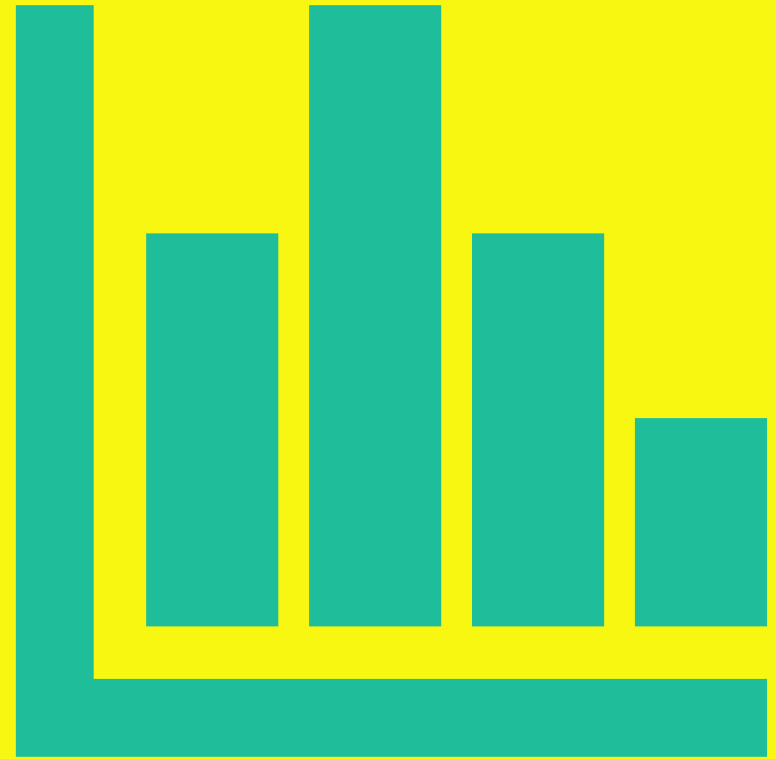
lesPeak Year of SalesSalesPerGameTop Genre KPIMissing Value Handling SheetTop Selling PlatformTop Selling RegionTop Selling



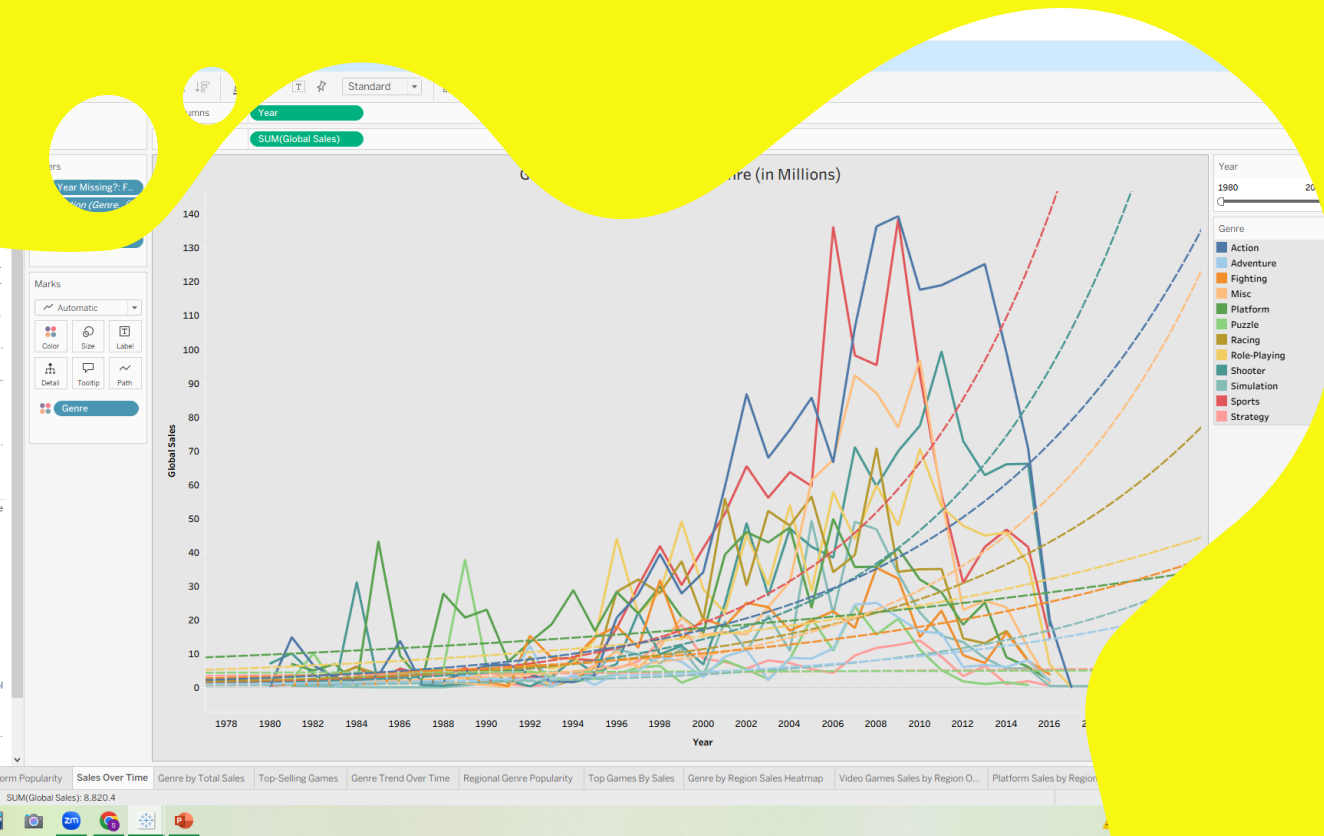
Executive Summary

- From 1980-2020, the video game industry generated over **8.8 billion units sold globally**.
- The **peak year was 2008** (678.9M units sold) – driven by *Wii* and *PlayStation 3*.
- **Average sales per game:** 0.5M units.
- **Top publisher:** Nintendo (1.7B units sold).
- **Summary Insight:**
The mid-2000s were a golden age for physical sales, but post-2010 the shift toward digital distribution disrupted traditional retail models.

Key Findings & Visualizations



Global Sales Over Time by Genre

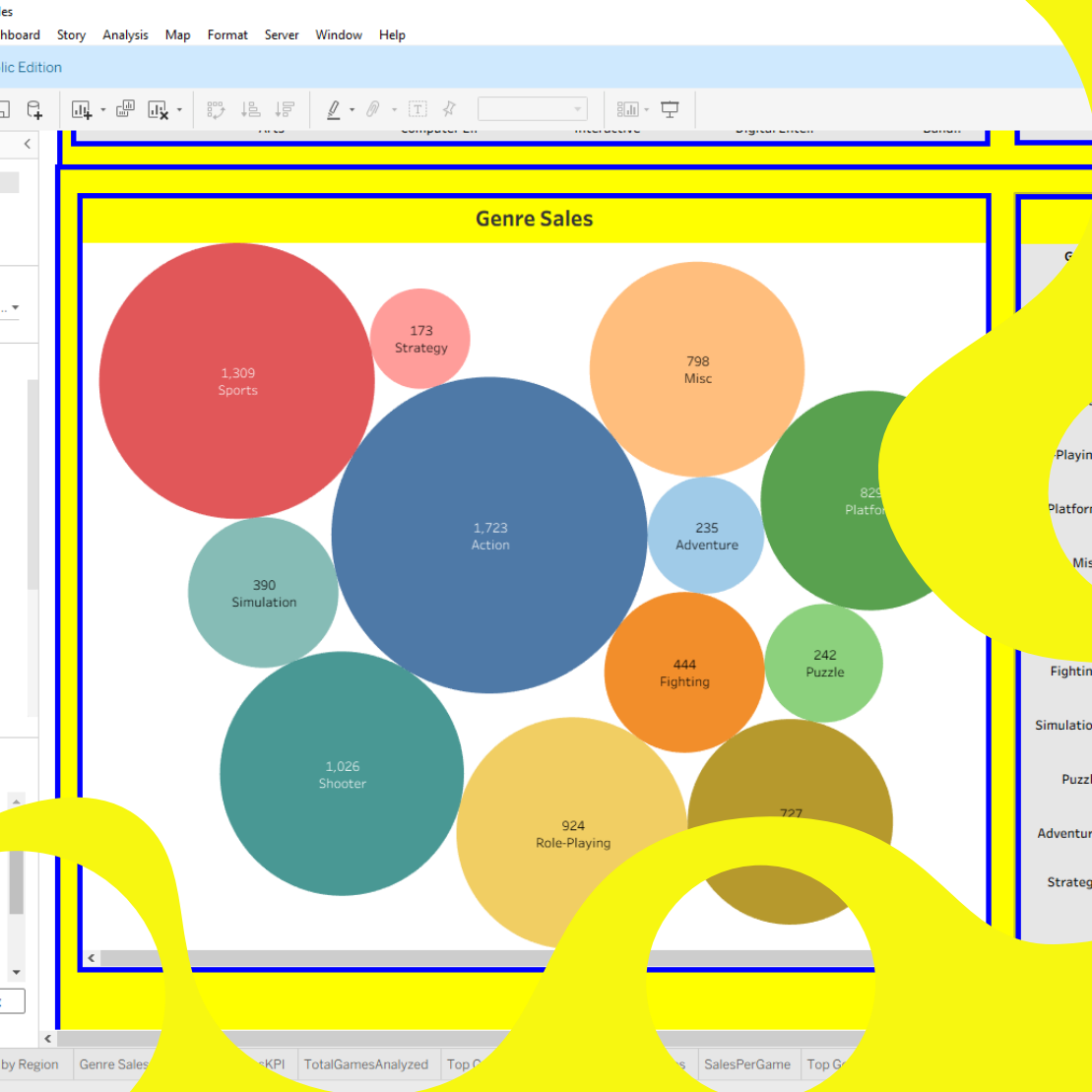


➤ Genre Concentration:

- Action, Sports, Shooter, and Role-Playing titles dominate about 75% of market volume.
- Indicates a highly consolidated demand space.

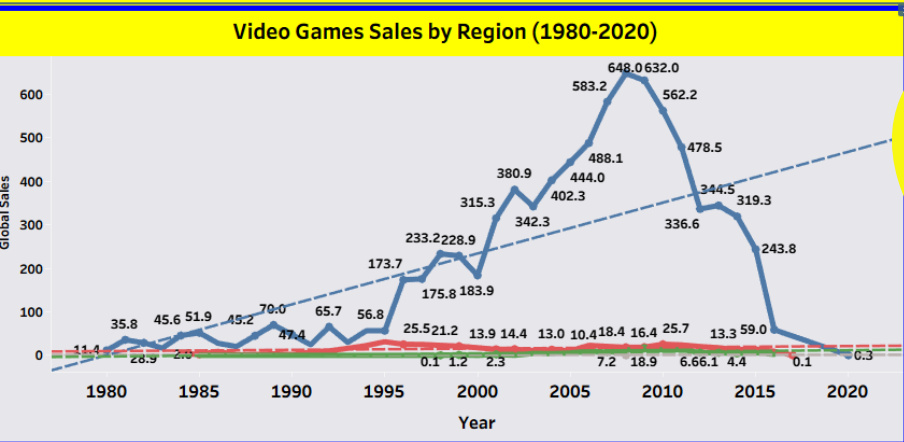
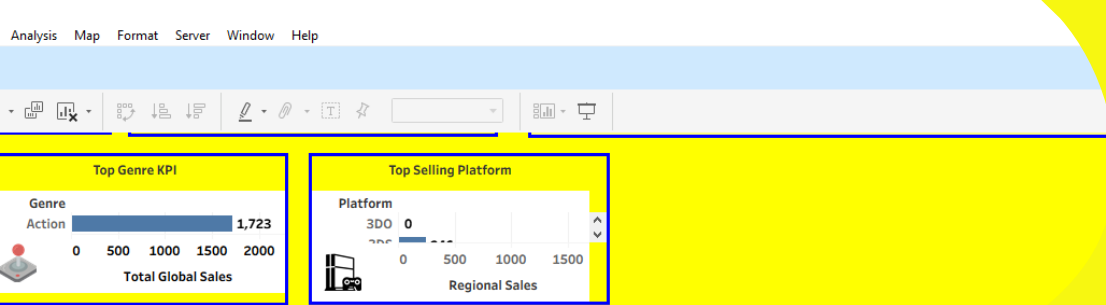
➤ Sales Plateau Post-2010:

- Retail sales fell sharply after 2010 as digital markets (Steam, Xbox Live, PSN) replaced physical distribution.



Genre Performance (1980-2020)

- **Action (1.7B)** and **Sports (1.3B)** dominate total global game sales during the time period.
- **Recommendation:**
 - **Publishers:** Prioritize multi-genre hybrids (Action + RPG, Sports + Simulation).
 - **Marketers:** Tailor campaigns to player motivations – competition and realism.
 - **Developers:** Leverage motion controls and immersive tech in these genres for retention.



“Video Games Sales by Region (1980–2020)”

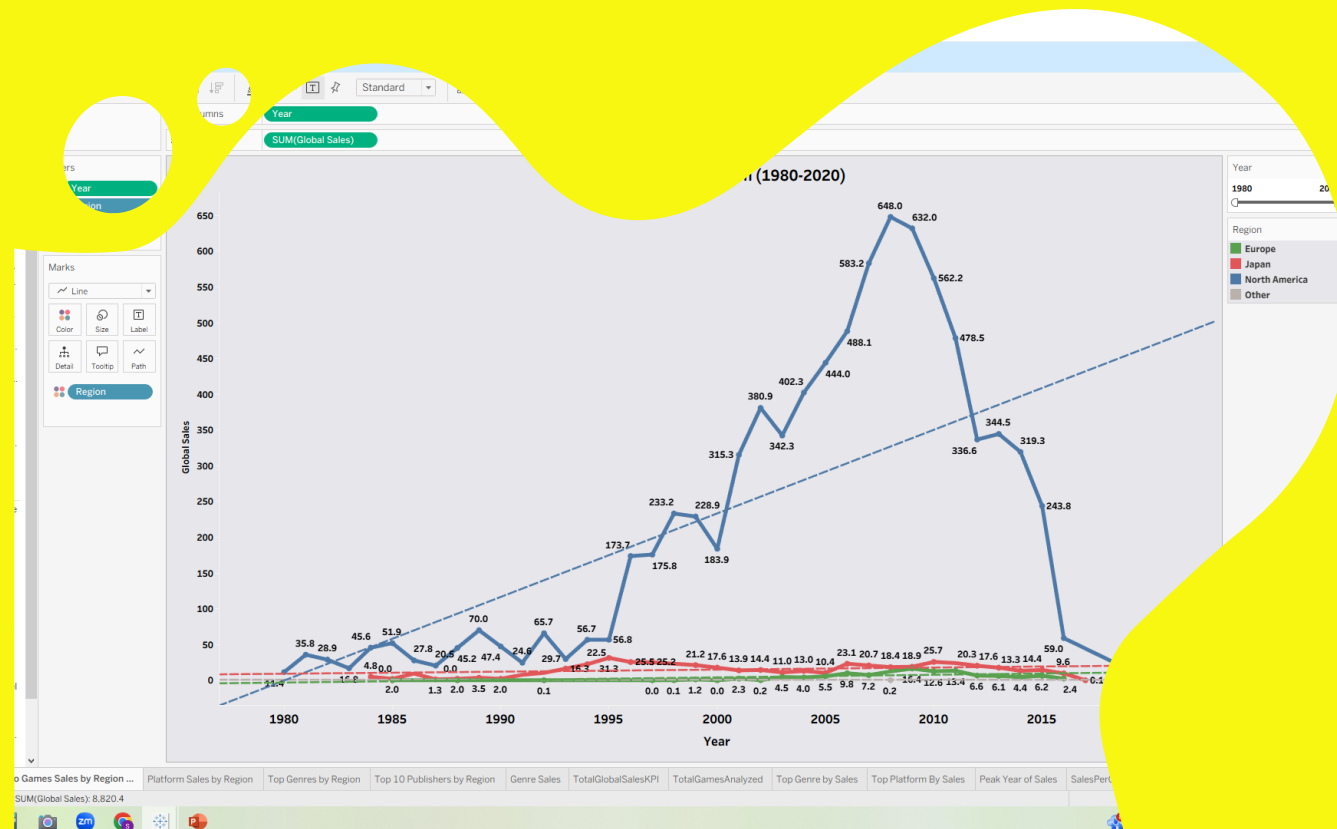
Findings

- **North America (blue)** - Remains the **dominant market**, showing explosive growth from the 1990s through its **peak of 648M in 2008**, followed by a steep decline as digital distribution replaced physical sales.
- **Japan (red)** - The **second strongest region**, maintaining **higher sales than Europe** for most years. Strong through the 1980s-1990s due to **Nintendo and PlayStation dominance**, peaking around **30-40M** before declining post-2005 as handheld and mobile gaming surged.
- **Europe (green)** - Shows **lower but steadier growth**, peaking around **20-25M**. Demonstrates greater **long-term stability** and smoother **digital adaptation** into the 2010s.
- **Other Regions (gray)** - Remain under **10M** until 2015, then show **gradual gains** as **mobile and cloud gaming** adoption increases in emerging markets.

Regional Trends (Based on Video Games Sales)

➤ Key Insights

- **Market Power Concentration:** The majority of global sales are concentrated in **North America and Japan**, which together drive over 80% of total physical game sales before 2010.
- **Platform Preferences:** Japan's market favored **portable and domestic consoles**, while North America dominated **home consoles** and franchise-based titles.
- **Digital Shift Impact:** Europe's flatter trendline shows it was **more adaptive to digital and downloadable content**, cushioning the decline after 2010.
- **Emerging Potential:** The slow upward curve for "Other" regions signals **early growth opportunities** tied to smartphone and low-cost streaming platforms.



Actionable Recommendations (Based on Regional Trends)

➤ For Publishers:

- **Launch Strategy:** Prioritize **North America and Japan** for major console releases; stagger European launches with localized versions to sustain sales beyond peak regions.
- **Distribution Model:** Expand **digital storefronts** (PlayStation Store, Steam, Switch eShop) to offset declines in physical media.
- **Franchise Repositioning:** Reintroduce legacy Japanese titles globally through remakes, remasters, and **cross-platform releases**.

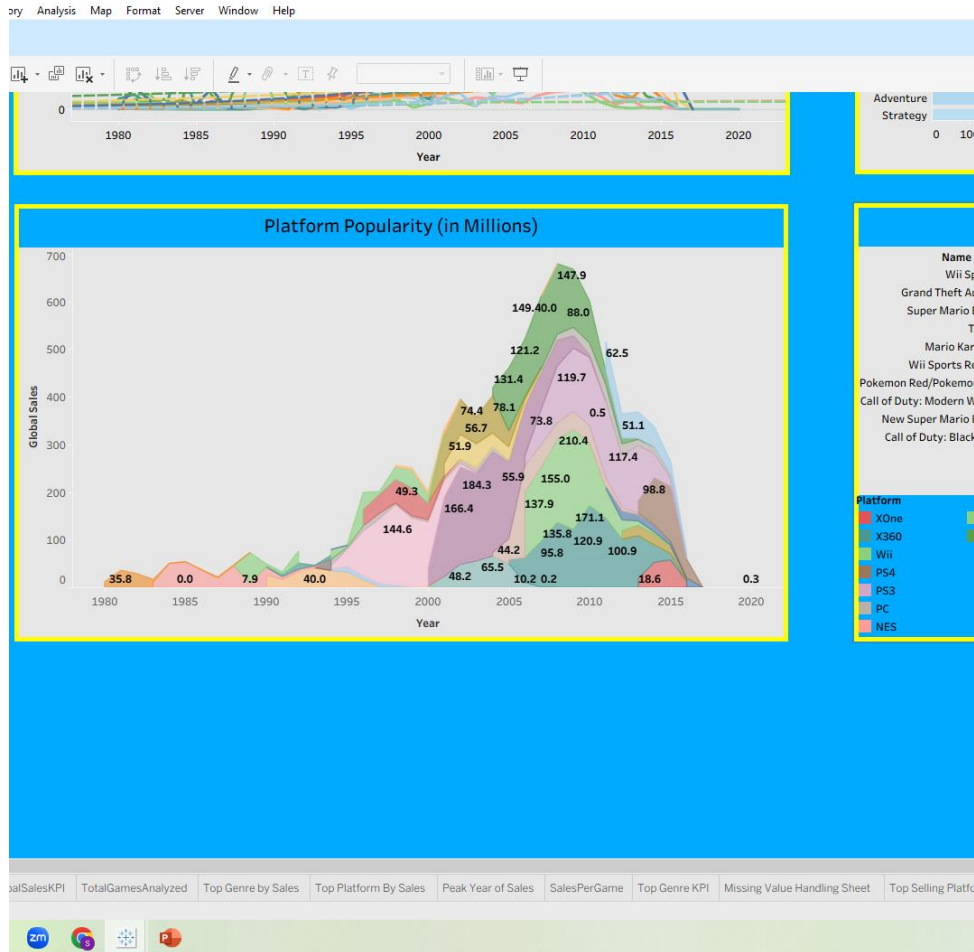
➤ For Developers:

- **Cultural Localization:** Tailor narratives and art styles for **Japanese and North American audiences** while adding **multilingual European versions**.
- **Platform Flexibility:** Develop engines that **support both consoles and mobile platforms** to reach diverse regions.
- **Innovation in Emerging Markets:** Create **mobile-first titles** optimized for low bandwidth and cloud streaming.

➤ For Policy & Market Analysts:

- **Invest in Digital Infrastructure:** Support policies that enhance **broadband and e-commerce access** in emerging markets.
- **Encourage Local Game Studios:** Offer **funding or tax incentives** to promote regional content creation.

Platform Popularity



Findings

- Total global sales **peaked around 2008-2009 (~700M)** – the height of the console era.
- **PS2, Wii, and DS** were top-selling platforms, each reaching **140M-150M** units.
- After 2010, sales **declined sharply** as gaming shifted to **digital and mobile** platforms.
- Console **life cycles shortened**, with faster turnover between generations.

Insights

- **PlayStation and Nintendo** dominate through brand loyalty and cross-generation continuity.
- **Innovation drives success:** motion controls (Wii), portability (DS), and media playback (PS2).
- **Digital transition** reshaped revenue streams, reducing reliance on physical game sales.

Actionable Recommendations (Based on Platform Popularity)

Publishers:

Prioritize **cross-platform releases** to reach wider audiences.

Remaster legacy hits for digital and streaming markets.

Developers:

Build **platform-flexible games** optimized for console + mobile.

Integrate **new tech** (VR, cloud play, motion input) to attract diverse players.

Industry Leaders:

Support **cloud gaming and online infrastructure** in emerging markets.

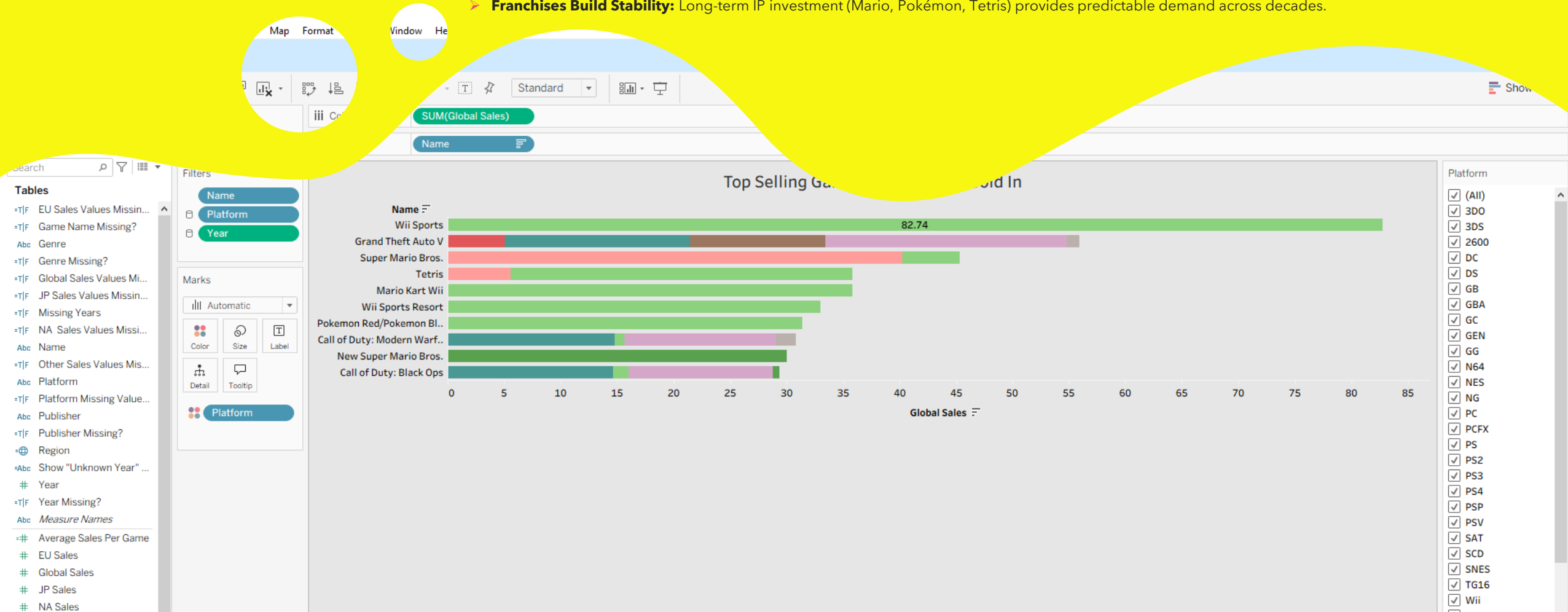
Top Selling Games

➤ Key Insights

- **Wii Sports (82.7 M)** – Dominates global sales largely because it was **bundled with most Wii consoles**, showing how hardware tie-ins can triple unit sales.
- **Grand Theft Auto V (≈ 55 M)** – Multi-platform success (PS3, PS4, Xbox, PC) demonstrates how **cross-platform releases extend a title's lifespan and total revenue**.
- **Super Mario Bros., Mario Kart Wii, Pokémon** – Franchise classics maintain **brand trust and loyalty**, proving that established IPs continue to outperform standalone games.

➤ Actionable Insights

- **Bundling Works:** Hardware-software packages like *Wii Sports* + *Wii Console* can turn average titles into global blockbusters.
- **Cross-Platform = Longevity:** Games released on multiple generations (e.g., *GTA V*) keep earning long after launch.
- **Franchises Build Stability:** Long-term IP investment (Mario, Pokémon, Tetris) provides predictable demand across decades.



Actionable Recommendations (based on Key Findings related to Top-Selling Games)

➤ Recommendations

➤ For Developers:

- Reuse proven **game engines** and extend franchises across generations to lower costs and boost consistency.
- Build games with **portability in mind** to support re-releases on future hardware.

➤ For Publishers:

- **Bundle flagship titles** with new consoles to accelerate adoption.
- Plan **multi-platform rollouts** from day one to capture wider markets.
- Pair new IPs with console bundles or promotional tie-ins for stronger early sales.

➤ For Marketing Teams:

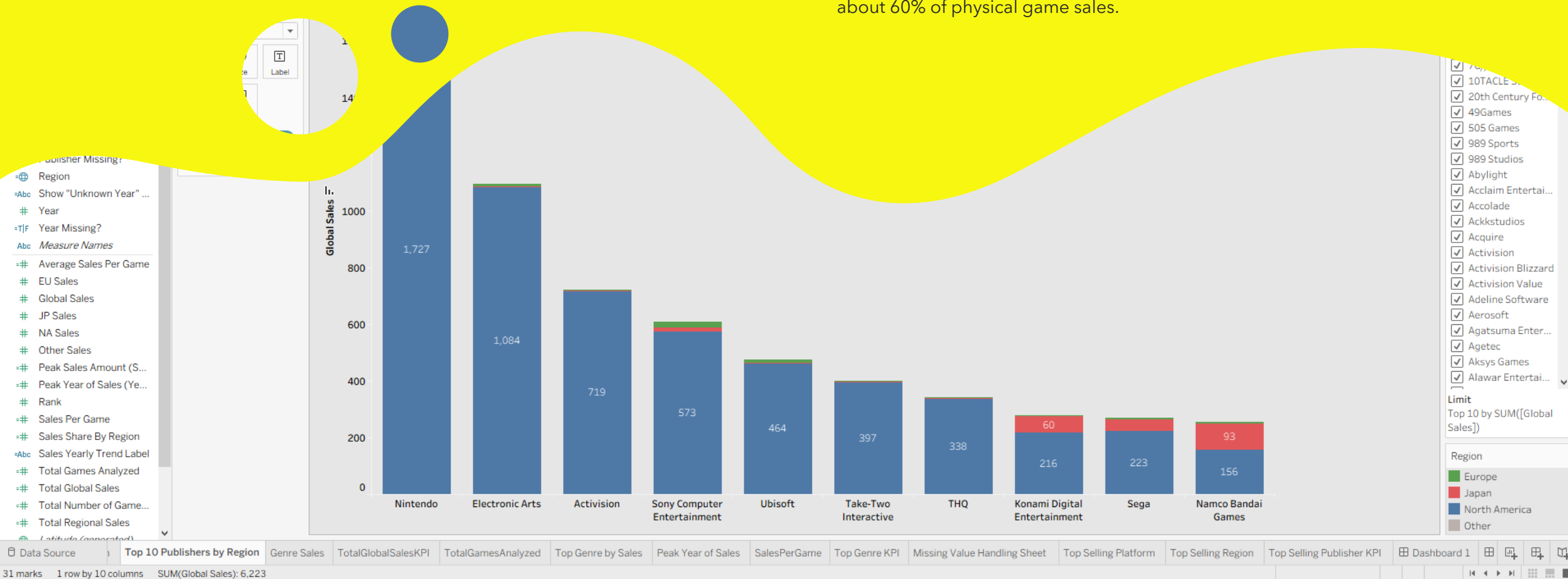
- Position bundles as **"instant-fun" starter packs** for new console owners.
- Re-market classic titles as **"remastered favorites"** for nostalgia buyers.
- Use regional data to target the **top-performing consoles and genres** per market.

Global Sales by Publisher

➤ **Nintendo** surpasses **Electronic Arts** and **Activision**.

➤ **Stakeholder Roles:**

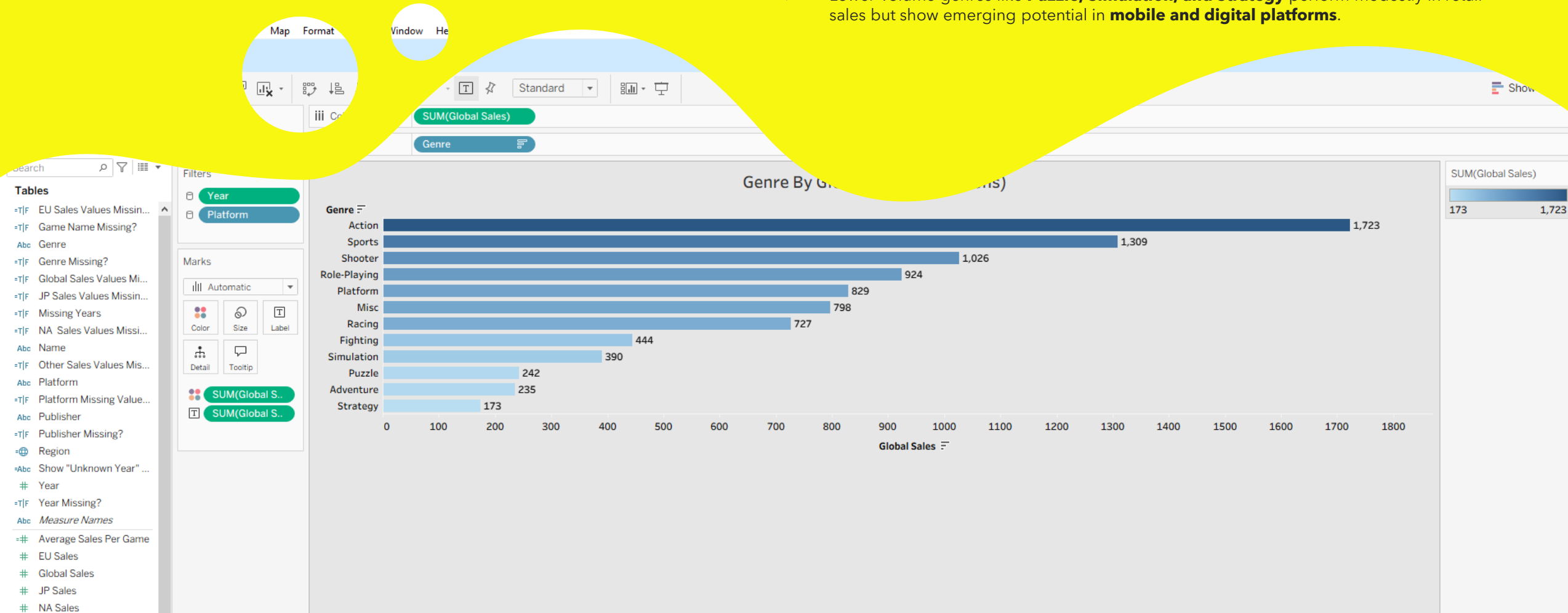
- **Publishers:** Use Nintendo's model by developing exclusive titles to strengthen brand ecosystems.
- **Retailers:** Feature brand-focused displays to capitalize on strong publisher recognition.
- **Analysts:** Monitor concentration ratios (HHI) – the top 5 publishers control about 60% of physical game sales.



Global Sales Performance by Genre (1980–2020)

Key Insights:

- **Action Games** lead globally with **1.7 million units sold**, followed by **Sports (1.3 B)** and **Shooter (0.9 M)**.
- These three genres account for roughly **75 % of all physical game sales** worldwide.
- **Role-Playing** and **Platform** titles remain resilient mid-tier performers, reflecting strong fan communities and legacy IPs.
- Lower-volume genres like **Puzzle, Simulation, and Strategy** perform modestly in retail sales but show emerging potential in **mobile and digital platforms**.

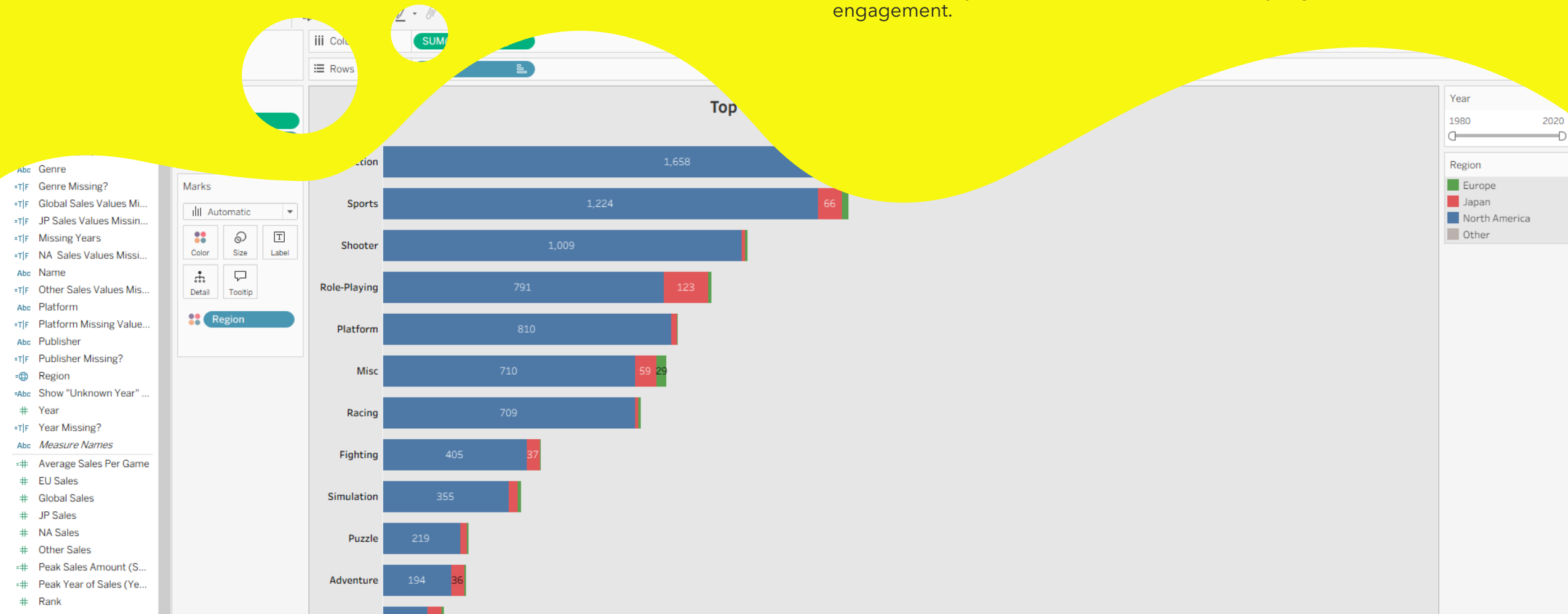


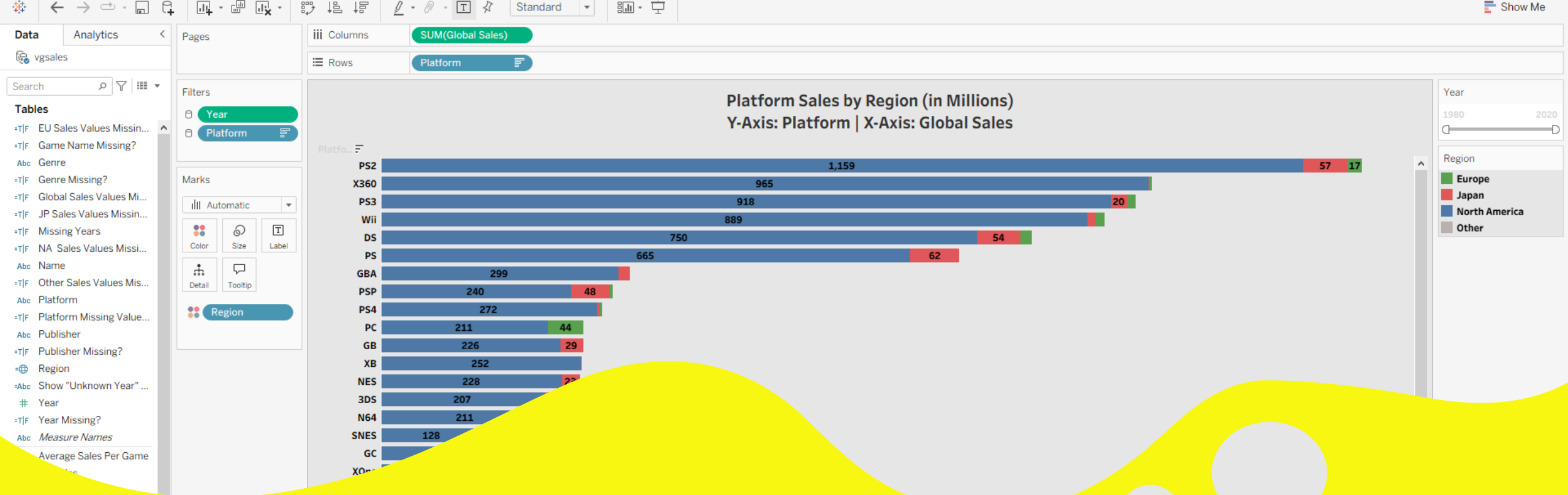
Actionable Insights

Stakeholder	Strategic Action	Rationale
Publishers (e.g., EA, Activision)	Reinforce annualized franchises in Action and Sports; develop cross-genre innovations like Action-RPG hybrids.	Proven sales consistency and cross-market appeal.
Developers	Leverage existing engines to produce genre spin-offs.	Reduces cost while diversifying catalog.
Marketers	Target region-specific demographics: Action/Sports (NA & EU), RPG (Japan).	Increases engagement and ad ROI.
Retailers	Bundle top genres with consoles and accessories.	Drives unit sales and pre-order volume.

Top Genres by Region (by Sales in Millions)

- Regional preferences reveal opportunities for localized content:
 - **Japan:** Prefers Role-Playing (JRPGs).
 - **North America & Europe:** Lead in Action, Shooter, and Sports.
- **Recommendation:**
 - **Developers:** Create culturally tailored game mechanics (story-driven JRPGs in Japan, competitive online play in NA/EU).
 - **Marketers:** Adjust visuals, ads, and voiceovers by region to maximize engagement.





Regional Hardware Strengths and Platform Market Share

- **PlayStation 2 (1,159 M units)** dominates globally, followed by **Xbox 360**, **PS3**, **Wii**, and **DS**.
- **North America** and **Europe** are the highest-volume regions, each driving over 70 % of console sales.
- **Japan** shows distinctive handheld and RPG-centric preferences (DS, PSP).
- Post-2010 declines reflect the industry's transition to digital platforms and mobile ecosystems.

Actionable Recommendations & Benefits

Stakeholder	Recommended Strategy	Benefit
Hardware Manufacturers (Sony, Nintendo, Microsoft)	Synchronize regional launch timelines and support local distributors.	Maximizes global adoption and reduces gray-market imports.
Publishers	Prioritize multi-platform releases (PlayStation + Xbox + Nintendo).	Increases total accessible market by 35-40 %.
Retailers	Stock higher volumes during the first 24 months of each console lifecycle.	Captures the peak revenue window.
Developers	Optimize for top-selling consoles before porting to niche platforms.	Ensures efficient resource allocation.
Policy Makers	Support regional manufacturing and logistics incentives.	Strengthens local gaming supply chains.

Methodology Overview



Data cleaning included:

Handling null values, removing duplicates (found no duplicates)
Creating calculated fields: *Global Sales*, *Peak Year*, *Avg. Sales per Game*, *Number of Games Analyzed*, etc.
Linking shared filters for Year, Genre, Platform, and Region.



Output: Two Tableau dashboards – one for global performance (blue) and another for detailed regional/genre breakdown (yellow).

Process Summary

- **Dataset Used:** Video Game Global Sales
- **Key Fields Reviewed:**
Name, Platform, Year, Genre, Publisher, NA_Sales, EU_Sales, JP_Sales, Other Sales, Global_Sales
- **Null Detection Method:**
Created Tableau calculated fields using ISNULL([Field]) to test each column for missing values.
- **Verification Results:**
 - Global_Sales and all regional sales fields contained **no nulls**
 - Year, Genre, Platform, and Publisher were **complete**
 - 3 Nulls were found in the Year field.
- **If Nulls Were Found:**
 - **Filter out** nulls in categorical fields
 - **Replace** nulls using:
 - IFNULL([Publisher], "Unknown Publisher") for text
- **Outcome:**
 - Dataset confirmed **clean and analysis-ready**
 - Ensured accurate KPIs and dashboards for *Regional & Genre Trends* and *Platform & Time Trends*

Data Preparation Methodology

Data Cleaning: Addressed missing or inconsistent values.

Feature Engineering: Derived *Top Genre, Top Platform, and Publisher KPIs*.

Visualization: Applied hierarchical filters for interactivity and storytelling flow.

Indicator Selection Criteria

- Indicators were chosen to balance business insight with technical accuracy:
 - *Total Global Sales* – market size indicator.
 - *Peak Year of Sales* – temporal trend indicator.
 - *Top Genre/Platform/Publisher* – performance indicator.
 - *Average Sales per Game* – profitability normalization metric.

Statistical Methods Used

- **The analysis is descriptive with diagnostic elements:**
 - Utilized **trend analysis** and **regression lines** for historical growth patterns.
 - Applied **aggregations** to evaluate genre dominance.
 - Integrated **interactive filters** to allow exploratory analysis.
- **Next Phase:** Add predictive modeling (ARIMA, Prophet) for 2030 forecasts.



Data Limitations

- Dataset excludes mobile and digital-only sales.
- Limited current-gen console data after 2016.
- Includes only titles selling over 100K copies – excludes indie markets.
- **Impact:** The analysis best represents physical console gaming, not digital-first economies.



Future Analysis Recommendations

- Integrate live APIs (Steam, Epic, Xbox) to capture digital sales.
- Build predictive dashboards for platform and genre forecasting.
- Add sentiment analysis (Metacritic, Reddit) to correlate player feedback with sales.
- Create real-time KPI alerts for publishers and analysts.



Visualization Recap & Insights Integration

- The **blue dashboard** gives a macro, executive-level view: total global performance, peaks, and cross-genre insights.
- The **yellow dashboard** delivers micro-level granularity: regional trends, publisher rankings, publisher ranking, and genre-specific dynamics.
- Together, these dashboards offer a unified decision-support tool adaptable for both executives and analysts.

Acknowledgments & References

References:

Kaggle Dataset: *Video Game Sales (VGChartz)*

Tableau Public Visualizations

Tools: Tableau, Excel

Author: Suzanne Leonard



Thank You!